

Tran Viet Khoa

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EDUCATION

National University of Singapore (NUS)

August 2024 – May 2028

Bachelor of Engineering in Computer Engineering

- **GPA:** 4.20/5.0
- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Computer Organization, Intro to Information Security.

WORK EXPERIENCE

AutoveX

February 2026 – Present

Full Stack Software Developer Intern

- Develop backend API features using **FastAPI** and **Django**, implementing dynamic filtering for flexible database retrieval.
- Automate API documentation via Django REST Framework **serializers**, ensuring synchronization between backend logic and developer references.
- Orchestrate containerized backend services using **Docker** within a production deployment stack to maintain environment parity.
- Tech Stack: *Python (Django, FastAPI), Redis, Docker, Nginx, HTML.*

PROJECTS

KForum [GitHub](#)

November 2025 – Present

- Developed a **high-concurrency** web forum utilizing **vertical scaling** and **load balancing** to ensure low-latency performance and scalability for a growing user base.
- Architected a containerized full-stack application using **Docker** and **AWS EC2**, designing a secure PostgreSQL schema and **JWT-based** authentication with **HTTP-only cookie** handling.
- Engineered a bi-directional messaging layer using Socket.IO and **Redis Pub/Sub** for real-time commenting.
- Tech Stack: *Go (Gin), React, TypeScript, PostgreSQL, Docker, AWS EC2, Redis, Socket.IO*

NUS Schedule Boss (Orbital 2025 Apollo 11) [Github](#)

May 2025 – July 2025

- Developed an **AI-integrated** module management platform to streamline academic planning through NLP and interactive data visualization.
- Integrated an OpenAI-powered chatbot with persistent **MongoDB** history to automate scheduling via NLP and aggregated **Reddit-based** course reviews for informed decision-making.
- Engineered a full-stack solution featuring a dynamic timetable visualizer and a contextual notification system to provide personalized real-time schedule reminders.
- Tech Stack: *React, Node.js, Express.js, MongoDB, OpenAI API, Supabase, Python*

LEADERSHIP EXPERIENCE

Vietnamese Youth Alliance in Singapore (VNYA)

October 2024 – September 2025

Web Developer, Logistics Coordinator

- Led "Singapore Study Dream 2024" seminar (150+ attendees), partnering with 8 top universities and connecting 50 mentees with 40 mentors.
- Coordinated Career Fair 2025 logistics, engaging 100+ Vietnamese students in Singapore.
- Built user-friendly landing pages for major events, improving participant engagement and streamlining event coordination processes.

ADDITIONAL SKILLS

Programming: Go, C++, Python, JavaScript (Intermediate); Java (Beginner)

Frameworks & Web: React, Node.js, Express, Django, FastAPI, HTML/CSS, Nginx

Cloud & Tools: Docker, AWS (EC2), Redis, PostgreSQL, MongoDB, Git

AI & Data: OpenAI API, TensorFlow, PyTorch, Pandas, NumPy, Matplotlib

Languages: Vietnamese (Native), English (Fluent), Mandarin (Intermediate)